

■ 1.4.2 Representing Formulae

You should be able to type most kinds of mathematical formulae into *Mathematica* in a form that closely resembles the standard mathematical notation. You can use any of the functions in *Mathematica* in a symbolic fashion.

Here is a more complicated formula. (This occurs in an approximation to π due to Ramanujan.) Make sure that you put parentheses in the right places when you type in a formula like this.

```
In[1]:= Sqrt[8]/9801 (4n)! (1103 + 26390 n) /
        (n!^4 396^4 (4n))
Out[1]= 
$$\frac{2 \sqrt{2} (4n)! (1103 + 26390 n)}{9801 396^4 n!^4}$$

```

Mathematica has a large repertoire of rules for transforming expressions that you type in. These include the standard rules of algebra, such as $x - x = 0$, together with much more sophisticated rules involving higher mathematical functions.

Mathematica uses standard rules of algebra to replace $(\sqrt{1+x})^4$ by $(1+x)^2$.

```
In[2]:= Sqrt[1 + x]^4
Out[2]= (1 + x)^2
```

Mathematica knows no rules for this expression, so it leaves the expression in the original form you gave.

```
In[3]:= Sqrt[1 + Cos[x]]
Out[3]= Sqrt[1 + Cos[x]]
```

The notion of transformation rules is a very general one. In fact, you can think of the whole of *Mathematica* as simply a system for applying a collection of transformation rules to many different kinds of expressions.

The general principle that *Mathematica* follows is simple to state. It takes any expression you input, and gets results by applying a succession of transformation rules, stopping when it knows no more transformation rules that can be applied.

Take any expression, and apply transformation rules until the result no longer changes

The fundamental principle of *Mathematica*.

Mathematica knows two rules which it can apply to this expression. The first one makes the transformation $\cos[x] - 2\cos[x] \rightarrow -\cos[x]$. The second rule puts the $\cos$ and $\sin$ terms into a “standard order”, in this case alphabetical.

```
In[4]:= Sin[x] + Cos[x] - 2 Cos[x]
Out[4]= -Cos[x] + Sin[x]
```

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